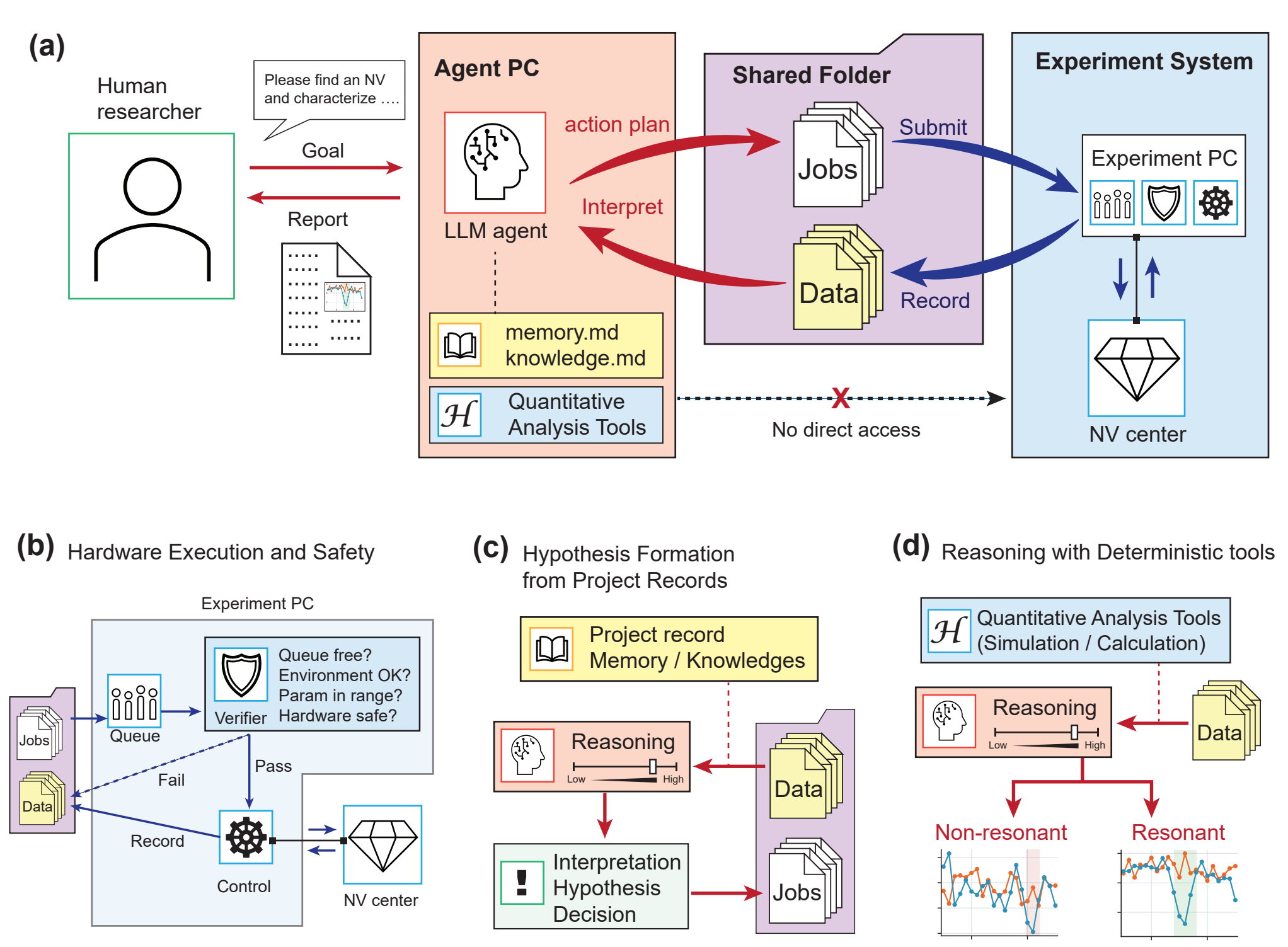

(a)
Human researcher
Please find an NV and characterize ….
Goal
Report
Agent PC
LLM agent
action plan
Interpret
memory.md
knowledge.md
Quantitative Analysis Tools
Shared Folder
Jobs
Data
Submit
Record
No direct access
Experiment System
Experiment PC
NV center
(b) Hardware Execution and Safety
Experiment PC
Jobs
Data
Queue
Verifier
Queue free?
Environment OK?
Param in range?
Hardware safe?
Fail
Pass
Record
Control
NV center
(c) Hypothesis Formation from Project Records
Project record
Memory / Knowledges
Reasoning
Low
High
Interpretation
Hypothesis
Decision
Data
Jobs
(d) Reasoning with Deterministic tools
Quantitative Analysis Tools
(Simulation / Calculation)
Reasoning
Low
High
Data
Non-resonant
Resonant